Daily AI Dev Digest

Wednesday, October 28, 2026

Build specialized multimodal models with LFM3

Source: Hugging Face Blog

The Brief

What shipped
Liquid AI announced the LFM3 family of specialized multimodal models, built from the LFM3-1.7B-Base backbone.

Capabilities
The family covers text, vision, audio, omni, and decision use cases; the text model has a 256K-token context window, and LFM3-VL-2B accepts video. Liquid AI reports LFM3-1.7B scores 38.42 versus 3.02 for LFM2.5-2.6B on WebArena-Verified, while LFM3-VL-2B averages 71.4 across 62 vision benchmarks versus 57.6 for LFM2.5-VL-3B; the audio encoder's time to first token is reported as low as 50 ms.

Availability
The article provides Hugging Face model IDs and Transformers loading instructions; the vision model requires transformers>=5.21.0. API access, pricing, regions, and model licenses are not stated.

Why it matters
Builders can evaluate separate modality-specific models rather than serving one general-purpose model, with the text model intended for local agents and the vision model supporting video input. The article says each model can be fine-tuned on a single GPU and describes edge deployment, but gives no comparative deployment cost figures.

Caveats
The performance figures are claims in the announcement; benchmark setup and independent validation are not provided in the supplied text. The article excerpt ends during the audio model section, and does not provide full details for all six models.

Full Announcement Details

Build specialized multimodal models with LFM3
LFM3 is our new generation of Liquid Foundation Models (LFMs), built to be the starting point for specialized intelligence. It brings text, vision, audio, omni, and decision models onto one shared backbone, all in the 2B parameter range. This means you can pick the model for your modality, fine-tune it on your own data on a single GPU, and deploy it where your application runs.
LFM3 brings four main changes compared to our previous LFM2 family:

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More capability per parameter: a new tri-hybrid architecture and training recipe allow LFM3-1.7B to outperform LFM2.5-2.6B while using 35% fewer parameters.
Every modality on one backbone: six 2B-class models (text, vision, audio, omni, and decision) trained from LFM3-1.7B-Base, including our first omni model and video input for our vision model.
Built for the edge: the text model fits in about 1 GB at 4-bit precision, new in-house vision and audio encoders speed up processing, and the text model's 256K context window lets agents run on-device.
Made to be specialized: every model fine-tunes on a single GPU, so you can adapt it to your data, terminology, and deployment environment.
How we built a more expressive backbone
LFM2 showed that a hybrid of short convolutions and attention can run fast on-device without sacrificing quality. For LFM3, we used our hardware-in-the-loop architecture search to redesign that architecture for agentic use cases, which need more capability per parameter and fast inference over long contexts. The search now covers 14 device and backend combinations, from phones and a Raspberry Pi 5 to laptops, cloud CPUs, GPUs, and the Qualcomm Hexagon NPU.
LFM3-1.7B-Base has 24 layers and four main changes:
Tri-hybrid layer mix: 13 gated short convolutions, 5 sliding-window attention (SWA) layers with a 256-token window, and 6 global attention layers. The convolutions use a kernel size of 7, up from 3 in LFM2.
Head-wise gated attention: a sigmoid gate scales each head's output, so a head can switch off when it has nothing to add. This can help suppress activation outliers that hurt quantization.
Partial RoPE: global attention layers rotate only the highest-frequency half of each head, so the other half matches on content regardless of distance. SWA layers keep full RoPE.
Sandwich norm: each block normalizes both its input and its output to keep activations well-scaled.
The payoff shows at long context. Per parameter, LFM3-1.7B matches LFM2-1.2B on short prompts across edge devices, and with 8K tokens in context it is about 25% faster. LFM3-1.7B-Base was pre-trained on 22T tokens with the same 128K vocabulary as LFM2.5-8B-A1B . The final 3.5T tokens are a mid-training stage on curated data with a larger share of agentic traces and tool calls, which also extends the context window to 256K tokens.
One backbone for six models
Every LFM3 model branches from LFM3-1.7B-Base. Instead of running one large general-purpose model, you pick the one that...

Want to learn more?

https://huggingface.co/blog/LiquidAI/lfm3

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